

# HelixVPN — Final Phase: General Availability (GA) Readiness

## HelixVPN — Final Phase: General Availability (GA) Readiness

**Revision:** 1 **Last modified:** 2026-07-04T12:00:00Z **Status:** draft specification — describes what to build/verify, not the shipping product.

**Scope of this document.** This is the **last phase before General Availability** — the hardening, launch-readiness, and go-to-market phase that sits **after** Phase 3's enterprise/scale features are built (docs/research/mvp3/MVP3\_ENTERPRISE\_SCALE.md) and after the 8-platform client suite (docs/research/mvp2/) and the control-plane MVP (docs/research/mvp/) are feature-complete. This document does **not** re-specify any feature — it specifies the **checklist, gates, and long-term operating model** that must be satisfied before HelixVPN is offered to the general public as a supported, paid product.

**Entry condition.** GA readiness work starts once every gate in docs/research/mvp3/MVP3\_ENTERPRISE\_SCALE.md §10.2 (G30–G37) is green, **and** the pre-existing Phase-3 "Extended Reach" gates already scoped in docs/research/mvp/final/09-phase3-reach-wbs.md (G20–G26 — HarmonyOS/Aurora native shims, WASM browser proxy, billing-optional foundation, third-party security audit, reproducible builds) are at least pass or an honestly-tracked pending\_device/ operator\_blocked state, per that document's own anti-bluff discipline. **This document does not duplicate the third-party security audit (HVPN-P3-E25) or reproducible-builds (HVPN-P3-E26) epics already scoped there** — it treats their outcomes as **inputs** to the GA launch checklist (§2.1, §2.2) and extends them to GA-

specific scope (load testing at commercial scale, chaos-testing the *enterprise* HA topology from Phase 3, not just the single-region Phase-2 topology).

**Why this phase exists as its own document.** "Feature-complete" and "ready to sell to the general public with a support contract" are different bars. A product can pass every functional gate in Phase 0-3 and still be unsafe to launch if there is no on-call rotation, no pen-test sign-off at the *current* (post-Phase-3) attack surface, no finalized ToS/DPA, or no plan for what happens to a paying customer's FFI-dependent integration when *helix-core*'s API changes. This document is that missing bar, made concrete and falsifiable.

---

## Table of contents

- [1. Executive summary](#)
  - [2. Production launch readiness checklist](#)
  - [3. Go-to-market readiness](#)
  - [4. Long-term maintenance model](#)
  - [5. Phased task/subtask WBS](#)
  - [6. Reconciliation with the existing audit/reproducible-builds epics](#)
  - [7. Decision register \(operator input needed\)](#)
  - [Sources](#)
- 

## 1. Executive summary

GA readiness has three cooperating strands, each with its own gate:

1. **Production launch readiness** (§2) — is the *system* safe and observably correct to run at commercial scale, under real failure conditions, with a real human on-call rotation behind it?
2. **Go-to-market readiness** (§3) — is the *business* ready: pricing, legal, distribution (8 app-store listings), marketing, and a working onboarding funnel?
3. **Long-term maintenance model** (§4) — once GA happens, what is the *contract* HelixVPN makes with its integrators and customers about versioning, deprecation, backward compatibility, and security-patch turnaround?

None of these three strands is "done" by writing documentation — each has a falsifiable acceptance criterion in the WBS (§5), in the same evidence-based style as every other phase's WBS in this project (mvp/final/06-phase0-spike-wbs.md, 07-phase1-mvp-wbs.md, 09-phase3-reach-wbs.md, mvp3/MVP3\_ENTERPRISE\_SCALE.md §10).

---

## 2. Production launch readiness checklist

### 2.1 Security audit / pen-test sign-off

mvp/final/09-phase3-reach-wbs.md's epic HVPN-P3-E25 already scopes a **third-party security audit** (crypto, control plane, shims, WASM proxy) with gate **G25** ("zero unresolved Critical/High"). GA readiness **extends** that engagement rather than repeating it:

- The Phase-3 audit (G25) covers the system as it existed at the *end of the platform-reach work* (HarmonyOS/Aurora/WASM proxy). By the time GA readiness starts, mvp3/'s enterprise surface (SAML/SCIM endpoints, the SCIM-to-identity bridge, the billing/payment adapter boundary, the multi-region control plane) did not exist yet when G25 closed.
- **GA-specific requirement: a follow-up penetration test scoped specifically to the new Phase-3 enterprise attack surface** — SSO/SAML assertion validation, SCIM endpoint authentication, the payment-adapter boundary (even though billing is off by default, the code path exists and must be tested), and the multi-region control-plane's inter-region trust boundary (does a compromised region-B helixd replica gain anything beyond region-B's own RLS-scoped tenants?).
- **Acceptance:** an independent report (same third-party auditor or a new one) with **zero unresolved Critical/High findings** against the Phase-3-complete system, remediation loop closed per the constitution's iterate-to-clean-GO discipline.  
UNCONFIRMED:/PENDING\_FORENSICS: is the honest state for any finding still in remediation — GA does not ship over an open Critical/High.

### 2.2 Load/performance testing at target scale

Extends the existing per-leg SLO budgets (data-plane throughput vs. bare-link, mvp/final/06-phase0-spike-wbs.md gate G1: "≥80% of bare-link throughput"; control-plane convergence, <1s p99) to **commercial scale**:

- Target scale is defined per subscription tier (mvp3/MVP3\_ENTERPRISE\_SCALE.md §8.1): the Enterprise tier's HA topology (§5 of that document) must sustain the concurrent-

device count and per-region throughput the sales/pricing model (§3.1 below) commits to, with headroom.

- Concrete load-test scenarios: (a) sustained enrollment burst (N devices enrolling concurrently — proves pki/ipam don't serialize badly under a real onboarding wave); (b) sustained WatchNetworkMap stream count per helixd replica at the HPA scale-out threshold (mvp3/ §5.3); (c) policy-recompile storm (many tenants' time-boundary conditions, mvp3/ §6.1, firing in the same minute) — proves the coordinator's diff-and-push stays inside the convergence SLO under a realistic worst case, not just a single-tenant test.
- **Acceptance:** captured p50/p95/p99 latency + throughput evidence at the committed scale, for every scenario above, with the SLO budget from mvp3/ §7.5 met or exceeded.

## 2.3 Chaos/failure-injection testing of the control plane and transport failover

Extends the chaos-test evidence already required by mvp/final/v06-deploy/ha-and-multiregion.md §8 (single-replica loss, Postgres failover, bus loss, gateway failover, split-brain) to the **Phase-3 multi-region topology with real regions, real HPA, and enterprise-pinned transport policy**:

- Re-run every chaos scenario from ha-and-multiregion.md §8 against the Phase-3 topology (§5 of mvp3/), not the single-region Phase-2 reference topology it was originally written against — the acceptance bar is unchanged (no tunnel drop on control-plane failure, bounded convergence pause, no silent corruption), but the evidence must be captured on the actual production-shaped deployment.
- New GA-specific chaos scenario: **transport-fallback-under-censorship simulation** — inject a DPI-style UDP block (as already done for gate G2 in Phase 0) against an **enterprise tenant configured to pin a transport with alert-not-fallback** (mvp3/ §9.5, HVPN-ENT-364) and confirm the alert fires and reaches the SIEM export path (mvp3/ §6.4) rather than the tunnel silently degrading its obfuscation posture.
- **Acceptance:** captured before/after transcripts for every scenario, no scenario producing a data-plane content loss (only bounded, honestly-scoped convergence pauses per the existing ha-and-multiregion.md §6 semantics).

## 2.4 Documentation completeness

Four audiences, each with a completeness bar:

| Audience  | Document set                     | Completeness bar                         |
|-----------|----------------------------------|------------------------------------------|
| End users | per-platform quick-start + FAQ + | every platform has a published guide; no |

| Audience        | Document set                                                                                                                               | Completeness bar                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                 | troubleshooting, for all 8 platforms in mvp2/MVP2_OVERVIEW.md §4                                                                           | "coming soon" placeholder pages                                                                                                    |
| Admins          | Console guide covering org/tenant/team setup, SSO/SCIM configuration, policy authoring, billing                                            | every Phase-3 admin-facing feature (mvp3/ §3–§9) has a corresponding admin-doc section                                             |
| API integrators | OpenAPI-generated REST reference + protobuf-generated agent-contract reference (schema-first, zero-drift per SPECIFICATION.md §4 clause 8) | reference docs are generated from the same schemas that generate the Dart/Go/Rust clients — never hand-written and liable to drift |
| Operations      | runbooks: region failover (mvp3/ §5.6/§10 HVPN-ENT-324), Postgres failover, key rotation, incident response (§2.5)                         | every chaos scenario in §2.3 has a matching runbook a human on-call engineer can execute without reading source code               |

## 2.5 Support / on-call rotation and incident-response process

- A defined on-call rotation (schedule mechanism is an operational decision, §7) with escalation policy tied to the SLA tiers already defined in mvp3/MVP3\_ENTERPRISE\_SCALE.md §7.5.
- An incident-response runbook: detect (via the observability pipeline, mvp3/ §7.1) → declare → communicate (status page, §2.6) → mitigate → postmortem — with the postmortem template requiring a root-cause statement in the constitution's no-guessing vocabulary (no "probably", "likely" — either a proven cause or an explicit UNCONFIRMED: with a follow-up tracked item).
- **Acceptance: a game-day exercise** — a real, scheduled incident simulation run against a staging deployment, on-call engineer pages, executes the runbook, writes the postmortem — with the full transcript captured as evidence.

## 2.6 Status page + public changelog

- Extends mvp3/MVP3\_ENTERPRISE\_SCALE.md §7.5's status-page mechanism (aggregating the existing /healthz signal) to a **public-facing** page with historical uptime + incident history.
- A public changelog is generated from the same tracked-item / release-tag discipline the project already uses internally (workable-items DB, §11.4.93) — filtered to user-visible entries

only (internal refactors, gate/test additions are not customer-facing changelog entries).

---

## 3. Go-to-market readiness

### 3.1 Pricing finalized

The subscription tiers are already schema-ready (mvp3/MVP3\_ENTERPRISE\_SCALE.md §8.1 — Self-Host Free / Team / Business / Enterprise, keyed on plan.max\_devices/max\_networks/max\_bytes\_mo).

**Exact price points, billing cadence (monthly vs. annual), and overage pricing are explicitly not decided by this document** — this is the single most consequential open decision blocking GA (§7, D-GA-1) and it gates the self-serve signup flow (§3.4) directly: the Console's signup UI cannot render a price it does not have.

### 3.2 Legal (ToS / privacy policy / DPA)

- **Terms of Service and Privacy Policy:** drafted against the architecture's actual data-handling reality — critically, the Privacy Policy can make an unusually strong, literally true claim ("**we cannot see what you connect to**") because that is a *build property* (S6, mvp/final/v05-security/audit-and-compliance.md §0), not a policy promise — legal review should be instructed to verify the policy text against the actual schema (no connections/traffic table exists) rather than draft generic VPN-industry boilerplate that overclaims or underclaims relative to what the architecture actually does.
- **Data Processing Agreement (DPA)** for enterprise/GDPR-scoped customers: must reference the concrete mechanisms in mvp3/MVP3\_ENTERPRISE\_SCALE.md §7.3's GDPR row (data minimisation by construction, the erasure runbook) rather than generic template language.
- **Acceptance:** legal counsel sign-off, with the sign-off record citing which architectural documents were reviewed (this is a documentation-completeness check, not a legal-content check this document is qualified to make).

### 3.3 App-store listings finalized for all 8 client platforms

Directly enumerates the platform matrix from mvp2/MVP2\_OVERVIEW.md §4.1:

| Platform                | Distribution channel                                                               | GA listing requirement                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| macOS                   | Mac App Store + direct .dmg (notarized)                                            | listing copy, screenshots, notarization, privacy-nutrition-label accuracy (must match the "we cannot see your traffic" claim)                                                 |
| Windows                 | Microsoft Store + direct .msi/.exe (code-signed)                                   | listing copy, screenshots, EV code-signing cert                                                                                                                               |
| Linux                   | .AppImage/.deb/.rpm direct download + distro repos where feasible                  | package repo listing (if pursued), reproducible-build attestation link (cross-ref HVPN-P3-E26)                                                                                |
| Android                 | Google Play + direct .apk                                                          | Play Data Safety form accuracy (again, must match the architectural privacy claim), listing copy                                                                              |
| iOS                     | Apple App Store                                                                    | App Privacy details, TestFlight beta completed before GA submission                                                                                                           |
| HarmonyOS               | Huawei AppGallery                                                                  | requires HVPN-P3-E21's HarmonyOS gate G21 green (device-proven tunnel) before a listing can honestly claim the app works                                                      |
| Aurora OS               | Aurora enterprise/government app channel (Russian-hosted, per mvp2/platform notes) | enterprise-SKU distribution process, requires HVPN-P3-E22's gate G22 green                                                                                                    |
| Web (browser extension) | Chrome Web Store, Firefox Add-ons, Edge Add-ons                                    | listing copy explicitly stating the browser-scoped-proxy limitation (mvp/final/09-phase3-reach-wbs.md HVPN-P3-232 — "not a system VPN") so the store listing never overclaims |

**Acceptance:** every row above has a submitted (not merely drafted) listing, and every listing's privacy/data-safety disclosure has been checked against the actual schema by an engineer, not copy-pasted from a competitor's listing.

## 3.4 Marketing site

Out of engineering scope beyond one hard requirement: **any technical claim on the marketing site must be traceable to an architectural document or a captured-evidence gate** in this project's specification set — the same anti-overclaim discipline as §3.2/§3.3. A marketing claim of "we don't log your traffic" is uniquely defensible here because it is architecturally true by construction (S6); a marketing claim should never be permitted to say more than that (e.g. "military-grade encryption" marketing fluff not tied to a concrete, cited mechanism is out of place next to a codebase this specific).

## 3.5 Customer onboarding flow

- **Self-serve** path: signup (§3.1 pricing-dependent) → SSO or anonymous-account creation (svc-identity.md §1.1, unchanged) → first-device enrollment wizard → connect. This is a Console/client-app UX deliverable, not a backend change — every backend call it makes already exists (CreateAccount/BeginOIDC/Enroll).
  - **Enterprise-sales-assisted** path: helixvpnctl org create (mvp3/MVP3\_ENTERPRISE\_SCALE.md §8.3) → SSO/SCIM configuration wizard → bulk device-enrollment-token distribution → admin Console walkthrough.
  - **Acceptance:** a first-time user (internal dogfood tester with no prior product knowledge) completes each path end-to-end without engineer assistance, timed, with friction points logged as tracked UX items — not a subjective "looks fine" sign-off.
- 

# 4. Long-term maintenance model

## 4.1 LTS / versioning policy

- HelixVPN adopts semantic versioning for every independently-releasable artifact: the control-plane API (helix-`proto-generated`), the client-core FFI (helix-`ffi`), and each client application's own release train (per-platform store versioning conventions layered on top of the shared semver).
- **LTS cadence:** a concrete cadence (e.g. "every 4th minor is an LTS line, supported N months") is a decision this document flags rather than invents (§7, D-GA-2) — but whatever cadence is chosen, it is published in the same place the public changelog (§2.6) lives, and every LTS line's end-of-support date is announced at least one full support-window in advance, never silently.



## 4.2 Deprecation policy

- Any protobuf field, REST endpoint, or FFI function marked for removal follows: **(1)** marked deprecated in the schema (protobuf [deprecated = true], OpenAPI deprecated: true, a Rust #[deprecated] attribute on the FFI symbol) in one release; **(2)** a minimum deprecation window (tied to the LTS cadence, §4.1) during which the old and new surface coexist; **(3)** removal only in a major version bump, never a minor/patch.
- This is a **direct extension** of the schema-first, zero-drift principle already in SPECIFICATION.md §4 clause 8 — because clients are *generated* from the schema, a deprecation is visible to every language binding simultaneously, with no risk of one platform's generated client silently drifting from another's.

## 4.3 Backward-compatibility guarantees

- **Client-core FFI** (helix-ffi, mvp/final/v04-client/ffi-surface.md): within an LTS line, the exported symbols (helix\_start, helix\_stop, helix\_status\_stream, etc., extended in Phase 3 with posture/conditional-access-aware fields per mvp3/§4.2) are additive-only — new optional fields, never a changed meaning or removed symbol. A third-party integrator building against helix-core at LTS version N can rebuild against any N.x without a source change.
- **Control-plane API**: the WatchNetworkMap protocol's version field and the Phase-3-added client\_protocol\_version negotiation (mvp3/MVP3\_ENTERPRISE\_SCALE.md §5.7) are the concrete mechanism that already makes this guarantee mechanically testable — a compatibility test suite runs an **old-version agent against a new-version coordinator** (and vice versa within the supported window) and asserts correct behavior, not just "the code compiles."
- **Acceptance**: a compatibility-matrix test (old client × new server, new client × old server, across the declared support window) is part of the standing regression suite, not a one-time manual check.

## 4.4 Security-patch SLA

| Severity                                                         | Patch SLA (time to fix released, from confirmed report) | Applies to                                       |
|------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|
| Critical (remote compromise, key-material exposure, auth bypass) | 72 hours                                                | control plane, helix-core, every client platform |
| High                                                             | 2 weeks                                                 | same                                             |
| Medium/Low                                                       |                                                         | same                                             |

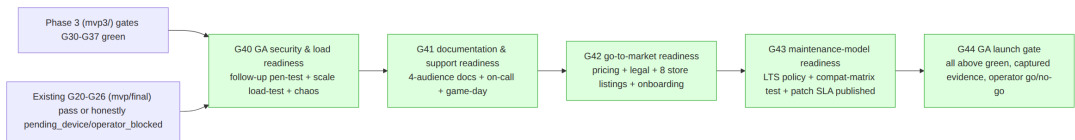
| Severity | Patch SLA (time to fix released, from confirmed report) | Applies to |
|----------|---------------------------------------------------------|------------|
|          | next scheduled minor release                            |            |

This table's exact numbers are a policy commitment the operator should confirm before publishing (§7, D-GA-3) — the structure (severity-tiered, time-bound, publicly stated) is the concrete deliverable; the specific hour/day counts are the negotiable part.

## 5. Phased task/subtask WBS

### 5.1 ID namespace

HVPN-GA-NNN, epics HVPN-GA-E4n, gates G40–G49 — a namespace distinct from both HVPN-P3-NNN/G20–G26 (existing platform-reach WBS) and HVPN-ENT-NNN/G30–G39 (mvp3/MVP3\_ENTERPRISE\_SCALE.md), so all three WBS sets can coexist in one docs/workable\_items.db (§11.4.93) without ID collision.



### 5.2 Epics

**HVPN-GA-E40 — Security & load readiness** epic · module: security,perf Gate **G40**. Delivers §2.1–§2.3.

- **HVPN-GA-400 — Follow-up penetration test scoped to the Phase-3 enterprise surface.** L(10) · deps: mvp3 G30–G37 green · tests: SEC Acceptance: independent report, zero unresolved Critical/High, remediation loop closed.
- **HVPN-GA-401 — Commercial-scale load test (enrollment burst, stream-count HPA threshold, policy-recompile storm).** L(9) · deps: HVPN-GA-400 · tests: PERF,STRESS Acceptance: captured p50/p95/p99 latency + throughput at committed scale, SLO met (mvp3/ §7.5).
- **HVPN-GA-402 — Re-run HA chaos suite against the Phase-3 multi-region topology.** L(8) · deps: HVPN-GA-401 · tests: CHAOS,REC
- **HVPN-GA-403 — Transport-fallback-under-censorship + pin-and-alert GA scenario.** M(5) · deps: HVPN-GA-402 · tests: CHAOS,SEC,REC

- **HVPN-GA-404 — G40 certification.** S(3) · deps: HVPN-GA-400..403 · tests: FA,CHAL

**HVPN-GA-E41 — Documentation & support readiness** epic · module: docs,ops Gate **G41**. Delivers §2.4–§2.6.

- **HVPN-GA-410 — Four-audience documentation completeness audit.** M(6) · deps: — · tests: FA Acceptance: the completeness table in §2.4 has zero missing rows; verified against the actual shipped feature set (mvp3/ §3–§9), not just a checklist self-report.
- **HVPN-GA-411 — On-call rotation + incident-response runbook + game-day exercise.** L(7) · deps: HVPN-GA-410 · tests: FA,REC Acceptance: captured game-day transcript (page → runbook execution → postmortem).
- **HVPN-GA-412 — Public status page + public changelog generator.** M(5) · deps: — · tests: UNIT,INT,FA
- **HVPN-GA-413 — G41 certification.** S(3) · deps: HVPN-GA-410..412 · tests: FA,CHAL

**HVPN-GA-E42 — Go-to-market readiness** epic · module: gtm,legal,console Gate **G42**. Delivers §3.

- **HVPN-GA-420 — Pricing finalized + Console signup pricing UI wired.** M(6) · deps: D-GA-1 resolved (§7) · tests: UI,UX,FA
- **HVPN-GA-421 — ToS/Privacy Policy/DPA drafted + legal sign-off citing architectural review.** M(5) · deps: — · tests: FA
- **HVPN-GA-422 — 8 app-store listings submitted, privacy disclosures verified against schema.** L(10) · deps: mvp/final G21,G22 green (HarmonyOS/Aurora) · tests: FA,REC
- **HVPN-GA-423 — Onboarding-flow dogfood test (self-serve + enterprise-assisted), timed, frictions logged.** M(6) · deps: HVPN-GA-420 · tests: UX,REC,FA
- **HVPN-GA-424 — G42 certification.** S(3) · deps: HVPN-GA-420..423 · tests: FA,CHAL

**HVPN-GA-E43 — Maintenance-model readiness** epic · module: helix-proto,helix-core Gate **G43**. Delivers §4.

- **HVPN-GA-430 — LTS/versioning policy published (cadence per D-GA-2, §7).** S(3) · deps: D-GA-2 resolved · tests: FA
- **HVPN-GA-431 — Deprecation-policy tooling (schema annotations + lint that blocks a non-major removal).** M(6) · deps: — · tests: UNIT,INT
- **HVPN-GA-432 — Backward-compatibility matrix test (old client x new server, new client x old server).** L(8) · deps: HVPN-GA-431 · tests: INT,E2E,FA Acceptance: added to the standing regression suite; a deliberately-introduced breaking

change in a test branch makes this test FAIL (paired-mutation proof it actually catches drift).

- **HVPN-GA-433 — Security-patch SLA published (numbers per D-GA-3, §7).** S(3) · deps: D-GA-3 resolved · tests: FA
- **HVPN-GA-434 — G43 certification.** S(3) · deps: HVPN-GA-430..433 · tests: FA, CHAL

**HVPN-GA-E44 — GA launch gate** epic · module: – Gate **G44**. All of G40–G43 green, full-suite retest from the Phase-3 baseline (constitution §11.4.40), operator explicit go/no-go decision recorded.

---

## 6. Reconciliation with the existing audit/reproducible-builds epics

To avoid any ambiguity about ownership:

| Item                                     | Owning epic                                              | What GA readiness adds                                                                                                                                                                                                                                  |
|------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Third-party security audit               | HVPN-P3-E25 (mvp/final/09-phase3-reach-wbs.md), gate G25 | GA adds a <b>follow-up, narrower</b> pen-test scoped to the enterprise surface G25 predates (§2.1) — it does not re-run or replace G25                                                                                                                  |
| Reproducible builds                      | HVPN-P3-E26 (same document), gate G26                    | GA <b>requires</b> G26 as an entry condition (public reproducible-build attestation is referenced from the Linux/.AppImage app-store listing, §3.3) but does not re-specify the reproducible-build mechanism itself                                     |
| Billing-optional multi-tenant foundation | HVPN-P3-E24 (same document)                              | Superseded in practice by the fuller organization/billing model in mvp3/MVP3_ENTERPRISE_SCALE.md §3, §8 — GA readiness's pricing/legal work (§3.1, §3.2) operates on the mvp3/ model, which is additive on top of E24's schema, not a replacement of it |

---

## 7. Decision register (operator input needed)

| #      | Decision                                                                                                                      | Options                                                                                                 | Recommendation                                                                                                                                                     | Resolved by                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| D-GA-1 | Exact pricing (§3.1)                                                                                                          | per-tier \$ amounts, monthly vs. annual, overage pricing                                                | not specified — the single highest-priority open business decision blocking GA                                                                                     | operator + go-to-market, before HVPN-GA-420 starts |
| D-GA-2 | LTS cadence (§4.1)                                                                                                            | e.g. "every 4th minor," calendar-based ("every 6 months"), or no formal LTS (rolling support only)      | recommend calendar-based (simpler to communicate publicly) but not decided here                                                                                    | operator, before HVPN-GA-430                       |
| D-GA-3 | Security-patch SLA numbers (§4.4)                                                                                             | the 72h/2wk/next-release table as drafted, or tighter/looser per legal & insurance requirements         | draft table proposed; confirm before publishing                                                                                                                    | operator + legal                                   |
| D-GA-4 | Which compliance certification (if any) to have completed <i>before</i> GA vs. pursued post-GA                                | SOC2 Type II before GA (slower, stronger enterprise signal) vs. after GA (faster launch, retrofit risk) | cross-references mvp3/MVP3_ENTERPRISE_SCALE.md D-ENT-3 — same underlying decision, restated here because it also gates the enterprise app-store/legal claims in §3 | operator                                           |
| D-GA-5 | Launch sequencing across the 8 platforms (§3.3) — simultaneous GA on all 8, or staged (Tier-1 first per mvp2/MVP2_OVERVIEW.md | simultaneous vs. staged                                                                                 | recommend staged, gated by mvp/final G21/G22 device-proof status (avoids the honest PENDING_DEVICE items in that WBS blocking the entire GA date)                  | operator                                           |

| # | Decision                                                                        | Options | Recommendation | Resolved by |
|---|---------------------------------------------------------------------------------|---------|----------------|-------------|
|   | §4.2 priority tiers, HarmonyOS/Aurora once G21/G22 are genuinely device-proven) |         |                |             |

## Sources

- docs/research/mvp3/MVP3\_ENTERPRISE\_SCALE.md — the immediately-preceding phase this document depends on; entry condition, org/billing/SLA schema this document's go-to-market section builds on.
- docs/research/mvp/final/09-phase3-reach-wbs.md — pre-existing third-party audit (E25/G25) and reproducible-builds (E26/G26) epics, reconciled in §6.
- docs/research/mvp/final/v06-deploy/ha-and-multiregion.md — chaos-test discipline extended to commercial scale in §2.3.
- docs/research/mvp/final/SPECIFICATION.md — schema-first/zero-drift principle (§4 clause 8), the basis for §4.2/§4.3's deprecation and compatibility guarantees.
- docs/research/mvp/final/v04-client/ffi-surface.md — the helix-ffi surface whose backward-compatibility guarantee is specified in §4.3.
- docs/research/mvp2/MVP2\_OVERVIEW.md — the 8-platform matrix and priority tiers enumerated in §3.3.
- docs/research/UNIFIED\_PHASE\_ROADMAP.md — full phase reconciliation, read alongside this document.

*End of Final Phase (GA Readiness) specification.*